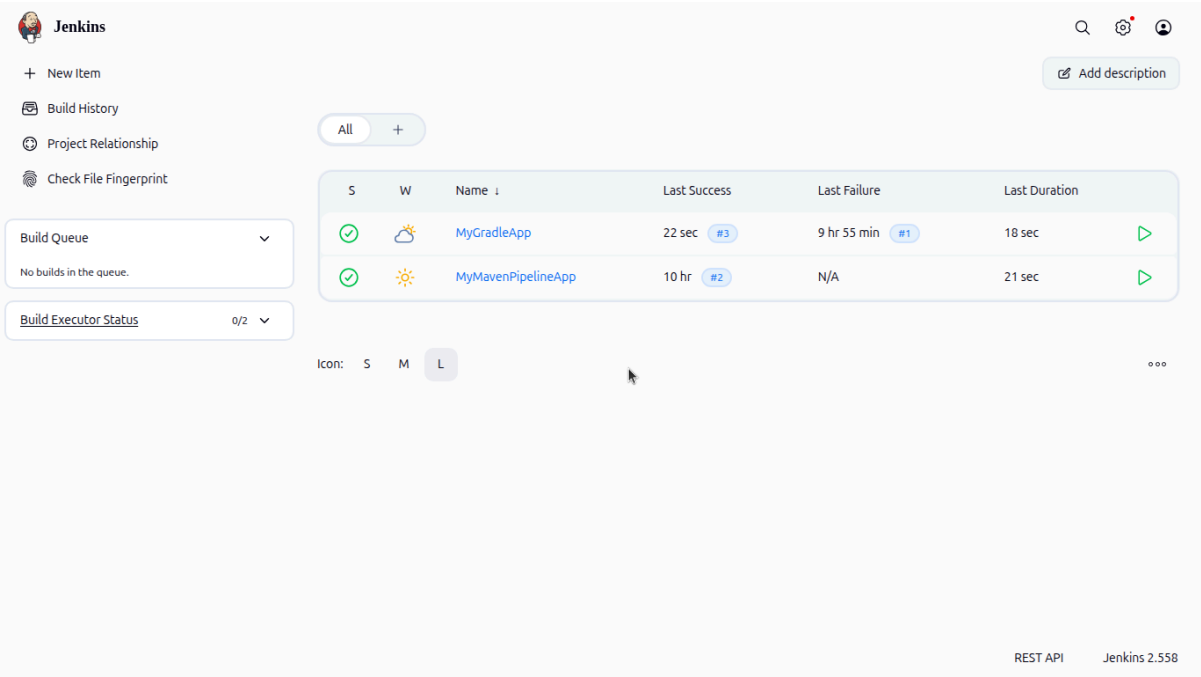
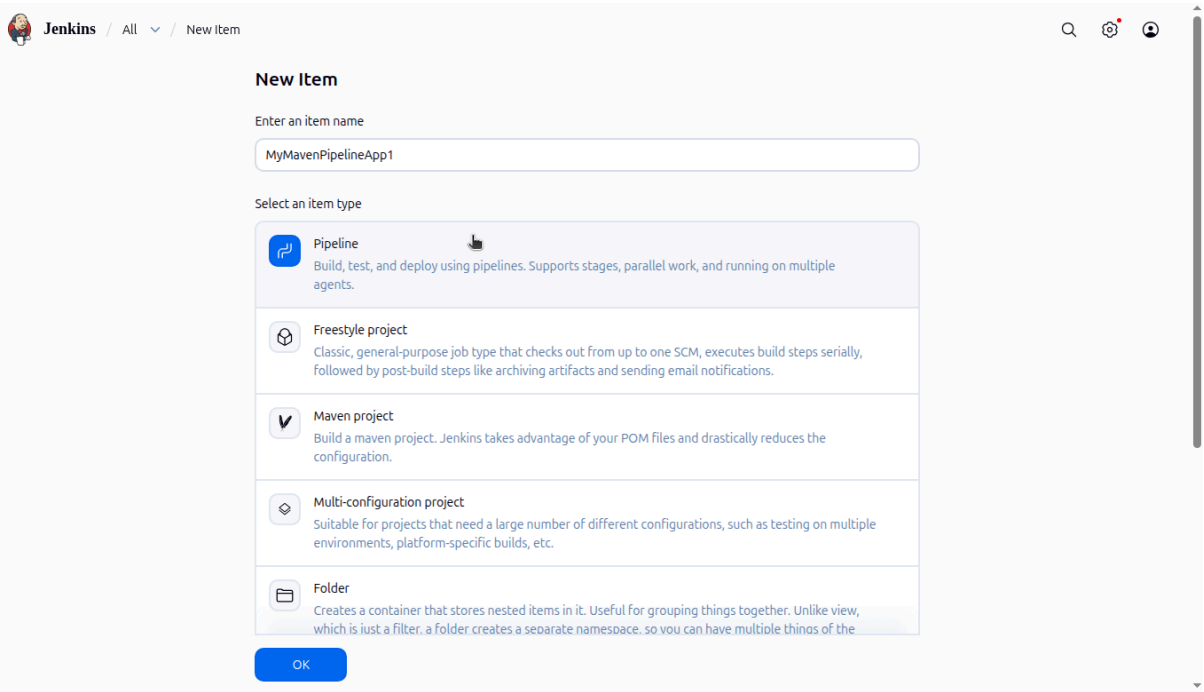
Jenkins Pipeline for MyGradleApp

Step 1: Open the Jenkins Dashboard in order to create a new item

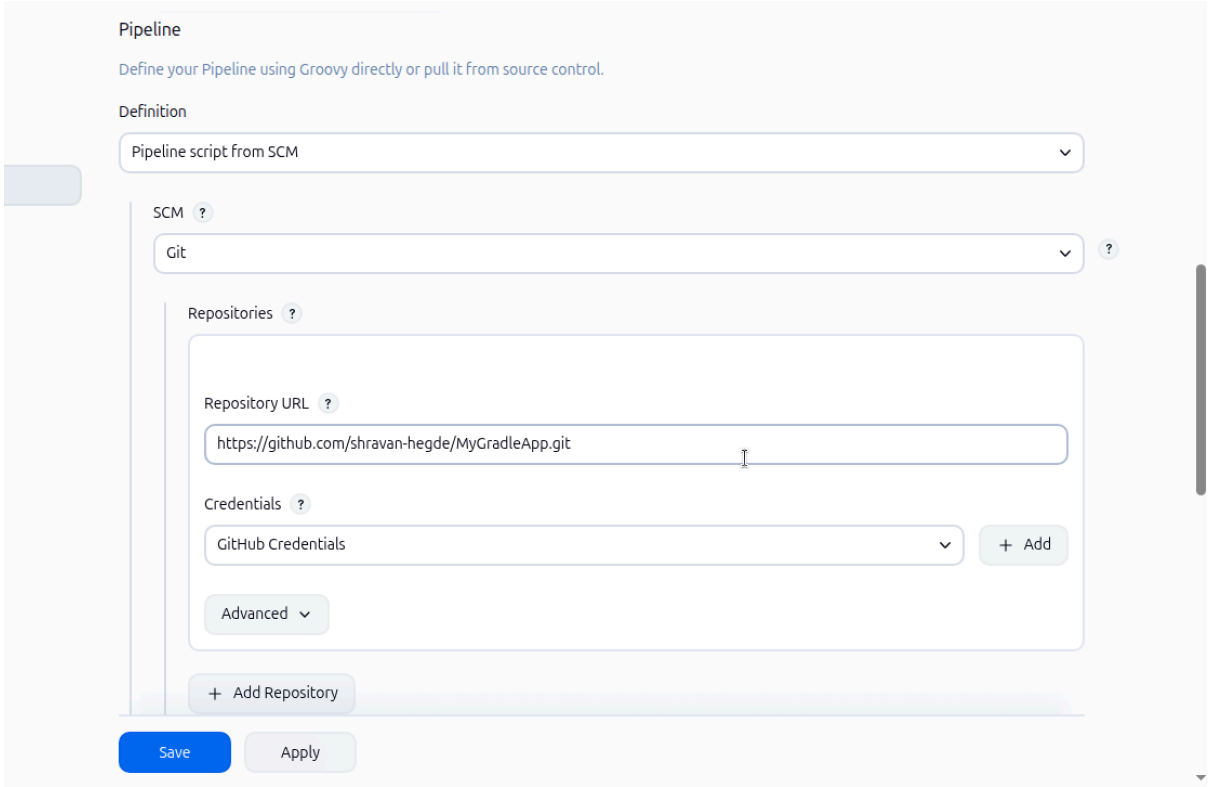


Step 2: Click on New Item in order for creating an New Item and thus give the name as "MyGradleApp" and Item type as Pipeline and

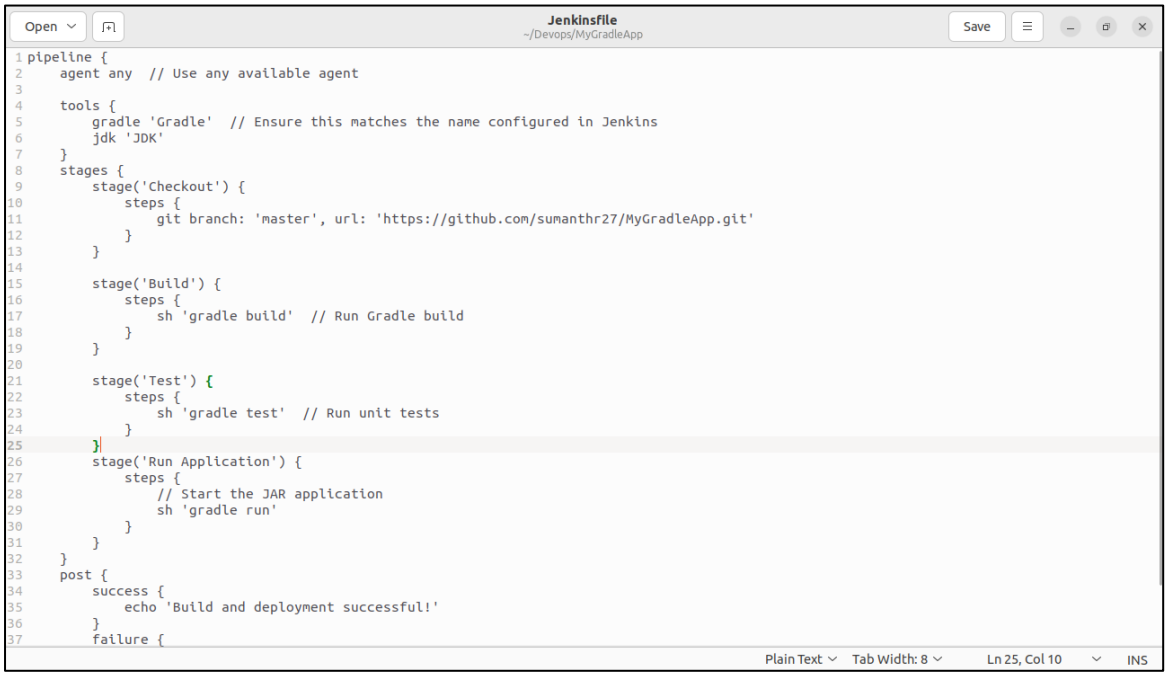
Create it



Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Paste the available code inside the Jenkinsfile

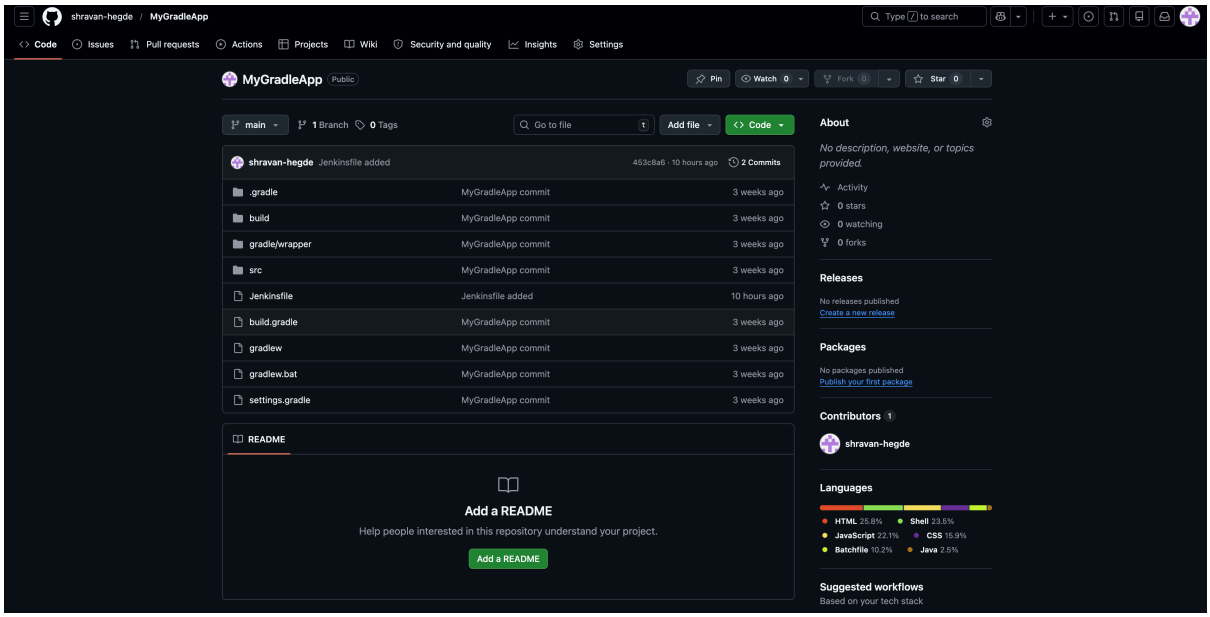



```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/sumanthr27/MyGradleApp.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }
    }
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
```

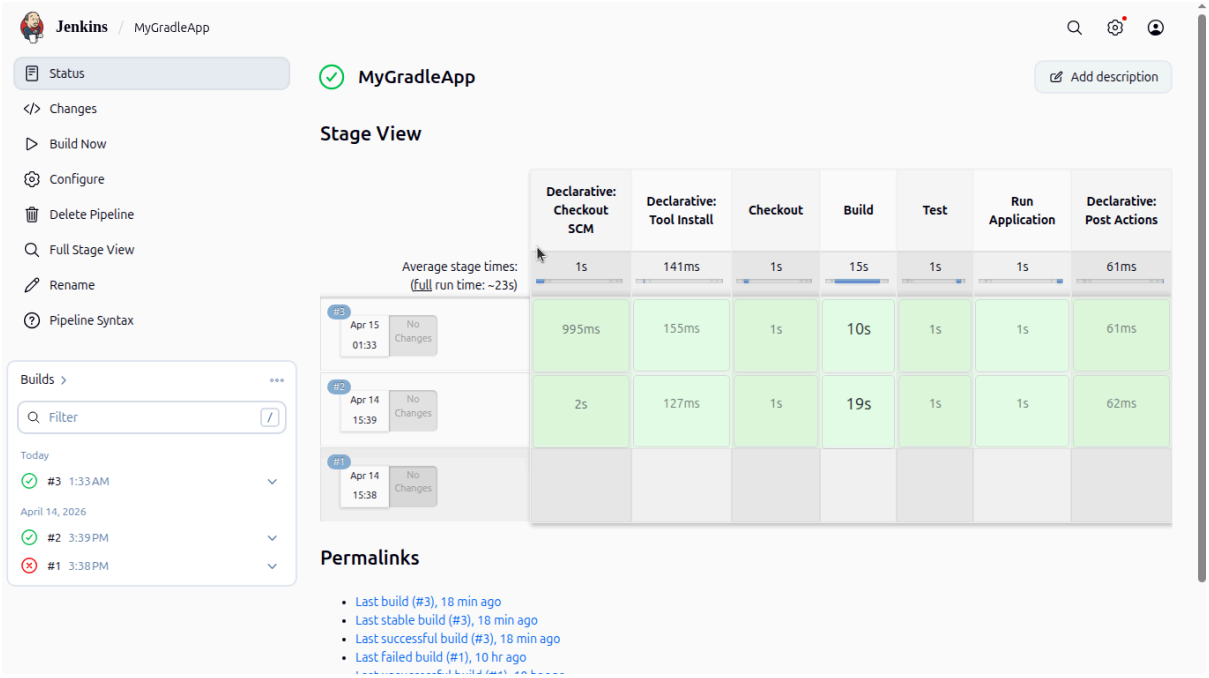
Step 5: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 6: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 7: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.


Jenkins / MyGradleApp
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Rename
Pipeline Syntax
Builds
Filter
Today
#3 1:33 AM
April 14, 2026
#2 3:39 PM
#1 3:38 PM
MyGradleApp
Add description
Stage View
Declarative: Checkout SCM
Declarative: Tool Install
Checkout
Build
Test
Run Application
Declarative: Post Actions
Average stage times:
(full run time: ~23s)
1s
141ms
1s
15s
1s
1s
61ms
#3 Apr 15 01:33 No Changes
995ms
155ms
1s
10s
1s
1s
61ms
#2 Apr 14 15:39 No Changes
2s
127ms
1s
19s
1s
1s
62ms
#1 Apr 14 15:38 No Changes
Permalinks
Last build (#3), 18 min ago
Last stable build (#3), 18 min ago
Last successful build (#3), 18 min ago
Last failed build (#1), 10 hr ago